This example also illustrates that looking only at the SWF's financial portfolio can result in the mistaken impression that risk aversion is changing. If the oil (or labor) income is bond-like, then reduced oil inflows imply that the SWF should reduce equities. An observer only looking at the financial portfolio might think the country has grown more risk averse. If oil income is stock-like, then the SWF increases equity holdings when oil inflows shrink. It may look as if the SWF has become more risk seeking, but all the SWF is doing is balancing oil risk across its life cycle.

Are You a Stock or a Bond?

A number of studies try to estimate the correlation between human capital returns and equities. At the overall level—a representative agent summing up the income of all workers in the United States or large subsets of workers—the correlation of human capital with stocks is low. Cocco, Gomes, and Maenhout (2005), for example, report that the correlation of labor income with stock returns is –2% for those with a college degree, 1% for those with only a high school education, and –1% for those not finishing high school. Thus for the average investor, human capital tends to be bond-like.[24] (Perhaps the one hundred minus your age rule is right after all, at least for the average person.)

The early life-cycle models relied on similar estimates.[25] Since they assumed the average worker had labor income that was bond-like with low correlations with equities, they predicted that young people should hold more equity than old people. They also predicted that young people should always hold some equities and poorer households would hold more equity than rich households. To understand the latter, bear in mind that, assuming they have the same risk profiles as rich households, the total wealth of poor families is dominated by human capital. Thus they would offset this bond-like capital by preferring equities in their financial portfolios.

It turns out all of these predictions are counterfactual.

3.5. WHAT PEOPLE ACTUALLY DO OVER THEIR LIFE CYCLES

There are three important facts of households' asset allocation over their life cycles:[26]

[24] See also Campbell et al. (2001). There is some variation, of course, in the estimates. Lustig and Van Nieuwerburgh (2008) are the exception in presenting a significantly negative correlation between labor income and stock returns. Bansal et al. (2011) estimate a large 35% correlation of labor income and equity returns. Neither paper uses direct estimates of labor income; they rely instead on asset prices to infer its behavior.

[25] These papers include Zeldes (1989), Heaton and Lucas (1997), Viceira (2001), and Cocco, Gomes, and Maenhout (2005).

[26] The first paper in this literature is Friend and Blume (1975). See also Heaton and Lucas (2000), Vissing-Jørgensen (2002), Ameriks and Zeldes (2004), and Calvet, Campbell, and Sodini (2006). Curcuru et al. (2004) is a literature summary.